



AJAY PRASATH P 2024-CSE ▾

A2**Started on** Wednesday, 17 September 2025, 11:06 AM**State** Finished**Completed on** Wednesday, 17 September 2025, 11:08 AM**Time taken** 1 min 58 secs**Marks** 1.00/1.00**Grade** 10.00 out of 10.00 (100%)

Question 1 | Correct | Mark 1.00 out of 1.00**Problem Statement:**

Given a sorted array and a value x, the floor of x is the largest element in array smaller than or equal to x. Write divide and conquer algorithm to find floor of x.

Input Format

First Line Contains Integer n – Size of array

Next n lines Contains n numbers – Elements of an array

Last Line Contains Integer x – Value for x

Output Format

First Line Contains Integer – Floor value for x

Answer: (penalty regime: 0 %)

```

1  #include <stdio.h>
2  int findFloor(int arr[], int low, int high, int x) {
3      if (low > high)
4          return -1;
5      int mid = (low + high) / 2;
6      if (arr[mid] == x)
7          return arr[mid];
8      if (arr[mid] > x)
9          return findFloor(arr, low, mid - 1, x);
10     int rightFloor = findFloor(arr, mid + 1, high, x);
11     return (rightFloor != -1) ? rightFloor : arr[mid];
12 }
13 int main() {
14     int n;
15     scanf("%d", &n);
16     int arr[n];
17     for (int i = 0; i < n; i++)
18         scanf("%d", &arr[i]);
19     int x;
20     scanf("%d", &x);
21     int floor = findFloor(arr, 0, n - 1, x);
22     printf("%d\n", floor);
23     return 0;
24 }
25

```

	Input	Expected	Got	
✓	6 1 2 8 10 12 19 5	2	2	✓
✓	5 10 22 85 108 129 100	85	85	✓

	Input	Expected	Got	
✓	7	9	9	✓
	3			
	5			
	7			
	9			
	11			
	13			
	15			
	10			

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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